



Alex Leung

Staff / Senior SWE • Tech Lead • Experimenter • Lifelong Student

University of Washington 2014 - 2018
Bachelor of Science in Computer Science
with an Entrepreneurship minor

at Flexport
in Bellevue
for 3.7 years
from Aug 2022
to Mar 2026

Tech Lead / Staff Software Engineer

Foundational Services `Next.js`, `TypeScript`, `GraphQL`, `Node.js`, `Bun`, `Effect TS`, `React Ink`, `Java`, `Envoy`, `Terraform`, `K8s`, `OAuth2`, `RBAC`, `AWS`, `Rust`, `Buildkite`

- Architected microfrontends framework for Flexport. Lead team of 7 engineers to create this, onboarded six separate orgs. Became foundation for Flexport's most profitable applications.
- Championed OE across GraphQL stack. Reduced latency by 50%. Increased reliability from 90% to 99% by architecting error attribution system, driving company-wide accountability.
- Standardized GraphQL authz and service-to-service authz across all microservices.
- Standardized service discovery on AWS CloudMap, merging several competing standards.
- Led team of 4 to consolidate ingress across all apps and APIs into unified API gateway.
- Drove consolidation of similar libraries and developer tools across codebases. Measured DevX improvements with velocity metrics, code reduction, and developer surveys.
- Led Flow to TypeScript migration for legacy websites and tools.
- Led migration from Algolia to OpenSearch for warehouse org, saving \$500k per year.

Tech Lead / Senior Software Engineer

Flex `TypeScript`, `GraphQL`, `Node.js`, `Express`, `Spinnaker`

- Drove adoption of UI-as-data microservice "Flex", enabling minimal device rendering code.
- Led migration of Flex from REST to GraphQL; integrated into Disney's supergraph.
- Partnered w/ overlapping Disney team. Flex took over UI as data, they took over i18n.

Software Development Engineer II

Fire TV `Java`, `Kotlin`, `Lambda`, `AWS`, `DynamoDB`

- Designed, implemented serverless component that ingests IMDB catalog. Event-driven.
- Last Mile Identity** `Java`, `AWS`, `Cognito`, `Coral (RPC)`, `Lambda`
- Extended OAuth microservice to support AuthN of various additional user types.
- Led redesign of RBAC authorization microservice.

Last Mile Account Management `Java`, `AWS`, `Coral (RPC)`, `Elasticsearch`, `React`

- Designed and implemented microservices as part of legacy monolith rearchitecture.
- Created hybrid web-app views in React for delivery driver onboarding.

Last Mile Risk and Compliance `Java`, `AWS`, `Angular`, `mTLS`, `Coral (RPC)`, `SQL`

- Integrated background check microservice with external vendor REST APIs.
- Extended services responsible for facial recognition of package deliverers
- Created data warehousing pipeline platform for Dynamo DB to Redshift streaming.

Payments Experience `Scala`, `Java`, `jQuery`, `SCSS`, `Coral (RPC)`

- Designed and implemented OTP verification for payment screens. Full-stack project.

Lead Software Engineer

`TypeScript`, `GraphQL`, `React`, `React Native`, `Ruby on Rails`, `Jenkins`, `AWS`, `MySQL`

- Built Ruby on Rails ERB / jQuery webpages then later migrated these to React + GraphQL.
- Designed MySQL schema. Optimized DB migrations, SQL transactions, ORM code.
- Maintained server infrastructure; configured EC2, Jenkins, Nginx, MySQL, Redis.
- Led project to create new mobile app using React Native, Redux, and Apollo GraphQL.
- Integrated 3rd-party APIs such as Stripe, Zillow, TransUnion, Twilio, Mailgun, etc.

Junior Software Engineer

`PHP`, `Laravel`, `Java`, `SCSS`, `jQuery`, `C#`, `Angular`, `Ionic`

- Created company marketing website and blog using CSS, jQuery, and PHP.
- Wrote web crawlers to pull online health-care data ontologies using C#.
- Wrote utilities to parse and normalize HL7 electronic health records using Java.
- Created proof of concept mobile application using Ionic 1.0.

at Hulu
Remotely
for 6 months
from Sep 2021
to Feb 2022

at Amazon
in Seattle
for 4.1 years
from Sep 2017
to Sep 2021

at Leasera
in Seattle
for 1.2 years
from Aug 2016
to Sep 2017

at ThruHealth
in Bellevue
for 1.9 years
from Aug 2014
to Jun 2016

📞 (425) 471-8377

✉️ alex.l.leung@gmail.com 🔗

🌐 alexleung.net 🔗

🐙 github.com/ozyman42 🔗

🌐 linkedin.com/in/alex-l-leung 🔗

OPEN SOURCE

- [Effect TS V4 API rework](#) 🔗
- [Apollo GraphQL subscriptions](#) 🔗
- [Deno QUIC \(HTTP3\) support](#) 🔗

TECHNICAL SKILLS

Maintainability: `SOLID`,
`GoF Design Patterns`,
`Service Oriented Arch.`,
`Event Driven Arch.`, `DRY`,
`Majestic Monoliths`, `Result<T,E>`,
`Dependency Injection`

Front-End: `Expo`, `TanStack`,
`Zustand`, `Jotai`, `Next.js`, `React`,
`React Native`, `Ionic`, `Angular`,
`jQuery`, `HTML`, `SCSS`, `Tailwind`,
`Webpack`, `Web Workers`, `Tamagui`

Server-Side: `Node.js`, `Kafka`,
`PostgreSQL`, `SQLite`, `Neo4j`,
`Cockroach DB`, `Redis`, `DynamoDB`,
`Serverless` (ex. `AWS Lambda`),
`Convex`, `SpacetimeDB`

Languages: `TypeScript`, `Effect TS`,
`Java`, `Kotlin`, `Scala`, `Rust`, `Ruby`,
`Solidity`, `Golang`, `Python`, `C#`, `C`,
`C++`, `PHP`, `SQL`

Dev Infrastructure: `AWS`, `GCP`,
`Github Actions`, `Buildkite`, `ArgoCD`,
`K8s`, `IaC`, `CDK`, `Terraform`, `Docker`,
`Linux`, `Envoy`, `Nginx`, `Apache`,
`Spinnaker`, `Backstage`

Transport/Security Protocols: `gRPC`,
`GraphQL`, `REST`, `mTLS`, `WebAuthN`,
`pRBAC`, `OAuth2`, `OIDC`

Web 3: `EVM (Solidity)`,
`Solana (Anchor)`

Observability: `OTel`, `Statsig`,
`Datadog`, `Grafana`, `Loki`, `Prometheus`